

ARGUS: Evidence-Grounded Auditing of Identification Assumptions in Climate-Policy Causal Evaluations

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Abstract

Difference-in-differences (DID) studies are widely used to evaluate climate policy, yet their identification assumptions are more often *asserted* than *audited*. Because the counterfactual is unobserved, we ask instead whether each assumption is adequately supported by the evidence a paper reports. We introduce ARGUS, an end-to-end, *bounded-agentic* auditing system in which an LLM runs behind a fixed, deterministic pipeline and scores DID studies on an 11-dimension assumption–implication–evidence rubric, retrieving evidence, judging its adequacy, and producing a risk report rather than a causal verdict. Under realistic flaw injection ARGUS catches all 11 injected flaws versus 5 for a same-model single-pass baseline, with no false alarms; across 27 real papers it exposes evidence grounding as the bottleneck, abstaining when support is absent; and a pilot diagnostic study reveals a systematic over-severity that a lightweight calibration layer partly corrects. ARGUS is thus a high-sensitivity screening tool that surfaces weakly evidenced identification assumptions for expert review.

1 Introduction

Difference-in-differences (DID) is widely used to evaluate climate and environmental policy, including carbon pricing, emissions-trading schemes, and emission standards. The credibility of these estimates depends on identification assumptions such as parallel trends, no anticipation, appropriate treatment of staggered adoption, and limited interference between units, each of which a decade of methodological work has shown to be fragile (Bertrand et al., 2004; Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021; Sun and Abraham, 2021; de Chaisemartin and D’Haultfoeuille, 2020; Roth et al., 2023). These assumptions are often difficult to assess in climate settings, where policies overlap, implementation is stag-

gered, spillovers cross administrative boundaries, and emissions outcomes are imperfectly measured (Appendix A).

Yet applied papers often assert these assumptions rather than audit whether the reported evidence supports them, one facet of a broader reliability gap in empirical research (Ioannidis, 2005; Silberzahn et al., 2018; Brodeur et al., 2020). A paper may state that parallel trends hold after presenting an event-study figure, while leaving the reader to determine whether the design, pre-period estimates, inference, and policy timing actually justify that claim. Performing this assessment manually is slow, expert-intensive, and difficult to standardize.

We argue that identification credibility is auditable even when the true causal effect is not, building on the research-design tradition in applied econometrics (Angrist and Pischke, 2010; Athey and Imbens, 2017; Rambachan and Roth, 2023). Because the counterfactual is unobserved, ARGUS does not ask whether an estimated effect is true. Instead, it asks whether each identifying assumption is adequately supported by the evidence reported in the paper, and where that support is weakest. ARGUS represents a DID study through an 11-dimension *assumption–implication–evidence* rubric and uses a bounded-agentic pipeline to retrieve relevant evidence, assess its adequacy, and produce an evidence-grounded risk report for expert review (Figure 1). Model agency is confined to scoped retrieval and assessment tasks behind a fixed, deterministic control flow, unlike free-roaming agents whose control flow is model-decided (Yao et al., 2023); ARGUS flags potential weaknesses rather than adjudicating causal truth.

Evaluating such an auditor is difficult because the true policy effect is unobserved. We therefore use *flaw injection* as local ground truth: starting from a supported study fixture, we introduce one

known identification flaw and test whether ARGUS detects it, remains quiet on the clean version, and localizes the affected dimension. We complement this controlled evaluation with deployment on 27 real DID papers and a small expert-labelled diagnostic pilot used to identify calibration and applicability failures.

Contributions. Our contributions are:

1. an 11-dimension framework that operationalizes DID identification credibility as auditable *assumption–implication–evidence* chains (§3);
2. a bounded-agentic, evidence-grounded auditing pipeline that separates retrieval uncertainty from substantive identification risk;
3. a layered evaluation combining realistic flaw injection, same-model architectural ablation, real-paper deployment, and a diagnostic expert-calibration pilot (§5).

2 Related Work

DID identification and its threats. A large econometrics literature formalizes when DID estimators fail under staggered timing and heterogeneous effects: the “forbidden” comparisons of two-way fixed effects (Goodman-Bacon, 2021), negative weights (de Chaisemartin and D’Haultfoeuille, 2020), robust estimators (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021), assumption and pre-trend surveys (Roth et al., 2023), and fragile inference under serial correlation (Bertrand et al., 2004). ARGUS adds no estimator: it *operationalizes* this literature as an auditable rubric, turning each known threat into a dimension whose supporting evidence can be checked in a given paper (§3).

LLM causal reasoning and benchmarks. A parallel line asks whether LLMs reason about causality. Benchmarks (CLadder (Jin et al., 2023), Corr2Cause (Jin et al., 2024), CausalBench (Zhou et al., 2024)) and critiques (Kiciman et al., 2024; Yang et al., 2024; Zečević et al., 2023) find models often reproduce causal *language* without reliable causal *inference*; a few move toward applied settings and toward separating *identification* from *estimation* (Lee et al., 2025; Sawarni et al., 2026). ARGUS asks the inverse question: are a *human* paper’s identification assumptions adequately evidenced? A causal verdict is out of scope by design.

Responsible AI for empirical research. AI is increasingly applied to research reliability: reproducibility checking, statistical-reporting validation, and AI-assisted review, motivated by wide analyst-driven result spreads (Silberzahn et al., 2018; Botvinik-Nezer et al., 2020) and specification searching in causal economics (Brodeur et al., 2020). ARGUS differs in target and posture: it audits *identification credibility* specifically (the part of an argument that licenses a causal claim) and issues no verdicts, producing evidence-grounded, human-reviewable reports that localize risk for the expert’s final call. A complementary line asks whether a model can *carry out* inference: running generated estimation code against synthetic data; ARGUS instead audits the prior layer, whether identification *assumptions* are supported by the evidence a paper reports.

Evaluation methodology and bounded agents. Our evaluation follows benchmarking lessons: holistic evaluation (Liang et al., 2023) over a single number (we report a layered retrieval/relevance/adequacy/risk profile); dynamic, adversarial benchmarking (Kiela et al., 2021) (our model-in-the-loop flaw injection); and the caution that operationalizing a capability is non-neutral (Raji et al., 2021; Gebru et al., 2021) (§6). Where ReAct-style agents (Yao et al., 2023) act over an open horizon, ARGUS *bounds* the paradigm (agency confined to two scoped sub-tasks behind deterministic control flow), trading autonomy for reproducibility and evidence-traceability.

3 The ARGUS Framework

3.1 Identification rubric

ARGUS decomposes a DID study into eleven auditable dimensions: parallel trends, no anticipation, staggered-timing handling, SUTVA/spillovers, control-group construction, specification, inference, sample-period selection, concurrent policies, placebo/robustness, and data measurement. Each dimension is expressed as an *assumption* (what must hold for identification), a testable *implication* (what should therefore be observable), and the *evidence* a credible paper would report to support it. For a carbon-pricing or emissions-trading evaluation, for instance, the parallel-trends dimension expects an event-study or pre-trend test with null pre-period coefficients, while no-anticipation

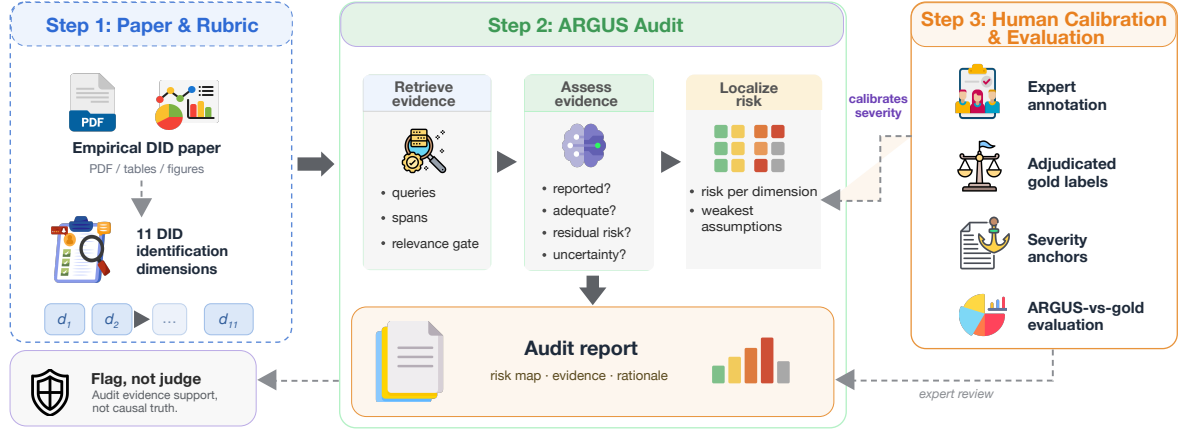


Figure 1: Overview of ARGUS. Step 1 converts an empirical DID paper into an auditable structure using an 11-dimension identification rubric. Step 2 retrieves dimension-specific evidence, assesses reported-evidence adequacy with a bounded LLM audit module, and localizes risk in a human-reviewable report. Step 3 uses expert annotation and adjudicated gold labels to evaluate and calibrate audit severity. ARGUS flags dimensions for expert review.

turns on the gap between a scheme’s *announcement* and *implementation* dates. The rubric is declarative, stored in a configuration file (`identification_dimensions.yaml`), so the audited dimensions are explicit and inspectable rather than buried in code. A companion flaw taxonomy (§4) maps each known identification threat one-to-one onto a dimension, which both documents the rubric’s coverage and supplies the perturbations used for evaluation.

3.2 Bounded-agentic pipeline

The audit runs a fixed, deterministic sequence, *decomposition* → *extraction* → *assessment* → *localization* → *report*, shown in the top band of Figure 2. The stage order is written in code and never chosen by a model. *Decomposition* instantiates the rubric for the paper at hand; *extraction* gathers, per dimension, the in-paper evidence relevant to that assumption; *assessment* evaluates each assumption–implication–evidence chain and emits a per-dimension risk judgement with a rationale and cited evidence; *localization* aggregates these judgements into a risk map ranking where support is weakest; and *report* renders a transparent, evidence-grounded document for a human reviewer.

Agency is confined to the two stages where open-ended judgement actually helps: extraction and assessment. Each invokes a bounded ReAct-style loop with three constraints: (i) a *small fixed tool set* passed in per call: `evidence_search` over the paper’s text, fig-

ures and tables, `figure_parse` for structured figure/table content, and `policy_lookup` for policy metadata such as announcement versus implementation dates; (ii) a *hard step budget* (`max_steps`) so a run cannot wander indefinitely; and (iii) a *logged trace* of every step, tool call, and cited piece of evidence. All other stages are plain orchestration with no agent loop.

Why bound the agency. A fully autonomous agent would make per-run behaviour path-dependent and high-variance, undermining the central evaluation claim of §4: that flaw injection provides *reproducible* local ground truth. Restricting agency to two clearly scoped sub-tasks preserves reproducibility (stable, comparable metrics) and evidence-traceability (every judgement anchored to logged evidence a human can inspect). The expert sits at the end of the pipeline: ARGUS surfaces and localizes identification risk; the judgement itself stays with the human.

3.3 Output

ARGUS emits (*study*, *identification dimension*, *risk*) triplets, each with an evidence-grounded rationale, alongside the human-readable report. It never produces a verdict on the true causal effect. This output shape keeps the system both auditable by a human and measurable by the evaluation that follows.

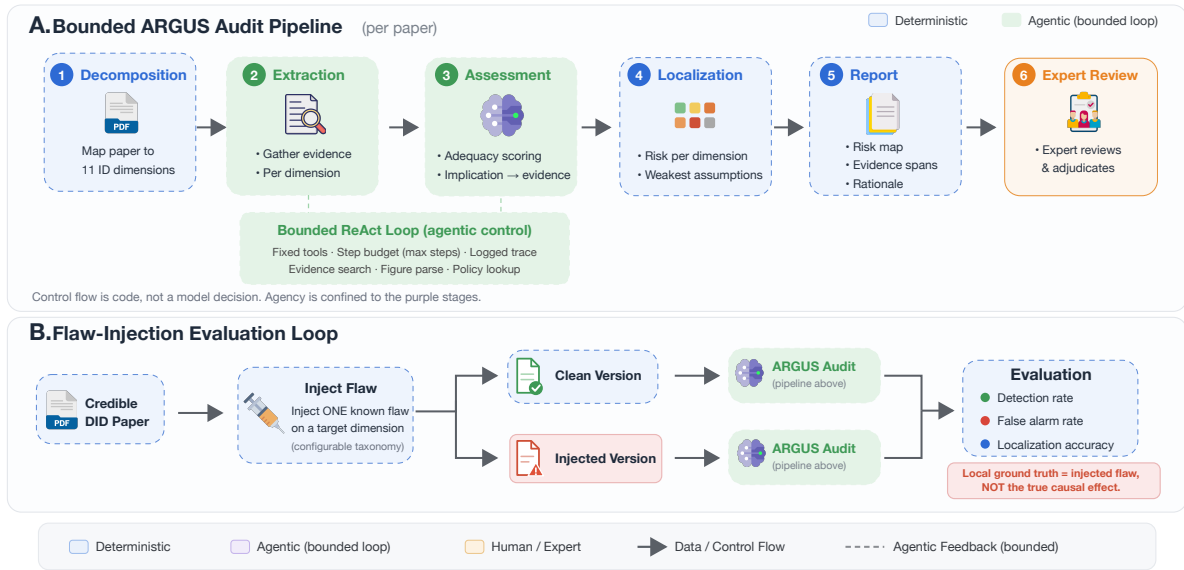


Figure 2: ARGUS architecture. The top band is the audit pipeline, a fixed deterministic sequence; agency is confined to the two agentic stages (extraction and assessment), each a bounded ReAct loop with a fixed tool set (evidence search, figure parse, policy lookup), a hard step budget, and a logged trace. The bottom band shows the flaw-injection evaluation loop, which injects one known flaw, audits the clean and injected versions, and scores detection, false alarm, and localization against the injected flaw as local ground truth, never against the unobserved true effect.

4 Evaluation by Flaw Injection

4.1 Local ground truth without the counterfactual

The defining difficulty in evaluating a causal-identification auditor is that the quantity of ultimate interest (the true causal effect of a carbon price, an emissions cap, or an emission standard) is never observed. We therefore do not attempt to certify that ARGUS recovers true effects. Instead we construct *local* ground truth by controlled perturbation. Beginning from a credible policy-evaluation paper, we inject exactly one known flaw that targets a single identification dimension, drawn from the flaw taxonomy (`flaw_taxonomy.yaml`). Because we control which dimension we damaged, we know the correct audit outcome for that dimension, without ever needing the true effect. The bottom band of Figure 2 shows the loop: the clean and injected versions are each audited, and three metrics are computed.

- **Detection:** does the injected version raise risk on the targeted dimension above a threshold?
- **False alarm:** does the clean version stay quiet on that dimension?

- **Localization:** is the targeted dimension the one the auditor flags most strongly?

4.2 Realistic perturbation, not sentinels

How a flaw is injected is a methodological choice that matters. A naïve injector appends a sentence that *names* the flaw: e.g. “treated and control have non-zero pre-period coefficients with significant leads.” If the detector then searches for exactly those phrases, detection becomes a tautology: the injector plants the very string the detector matches on. In our initial implementation this produced a perfect-looking but meaningless detection rate of 1.000.

We replace this with *realistic structural perturbation* of two kinds. *Omission* flaws remove the section, figure, or table that supplies the support for a dimension (e.g. deleting the placebo-robustness section or the balance table), modelling a paper that simply omits the check; an auditor can flag this non-circularly by noticing the *absence* of evidence. *Commission* flaws *rewrite* a section in flawed-but-natural prose, the way a real flawed paper would describe a mistaken design choice (e.g. pooling all province-years in a two-way fixed-effects regression under staggered policy timing), *without* containing any of the phrases the detector looks for. We enforce the latter with a regres-

sion test that fails if an injected paper contains the target dimension’s negative-signal phrases, so the circularity cannot silently return. Detection is thereby a genuine task rather than a string match.

5 Results

We evaluate ARGUS along four axes: a synthetic flaw-injection protocol, an architecture ablation, a run over 27 published difference-in-differences (DID) papers that stress-test the audit architecture across empirical economics, beyond the climate-policy setting (carbon pricing, emissions trading, emission standards) that defines its target deployment, and a calibrated human-gold pilot. Four findings emerge: (i) judging evidence *adequacy*, not keyword presence, is what detects identification flaws; (ii) the bounded architecture, not raw model capability, recovers flaws a single pass misses and turns retrieval failure into an auditable unknown signal; (iii) on real papers the binding constraint shifts from causal reasoning to evidence grounding; and (iv) against a human gold ARGUS is systematically over-severe, a bias that a lightweight calibration layer reduces without lowering high-risk detection on the pilot’s few high cells.

5.1 Adequacy beats presence; architecture beats a single pass

We compare three assessors under one flaw-injection protocol on the `clean_supported` fixture (one flaw injected per run, all eleven flaws; `max_steps=1`). A deterministic **keyword baseline** scores each dimension from the presence of supporting and risk-indicating vocabulary. The **two-stage ARGUS assessor** judges evidence *adequacy* (whether the reported evidence actually establishes the assumption) over the *same retrieved evidence* as the keyword baseline, so that pair isolates the judgement policy. A **single-pass** ablation removes the architecture entirely: the full paper and all eleven dimensions in one `gpt-4o` call, with the same model, temperature, risk schema, and scoring harness, but no staged retrieval, relevance gating, bounded tool use, or explicit unknown state. The only difference is architectural. Table 1 reports the headline metrics.

The keyword baseline is blind to commission flaws. The baseline never raises a false alarm and correctly flags the two omission flaws (`measurement_break`,

Table 1: Synthetic flaw injection (`clean_supported`, eleven flaws; all LLM rows `gpt-4o`): keyword baseline, a same-model *single-pass* ablation, and the *two-stage* ARGUS assessor. Keyword and two-stage share the same retrieved evidence; single-pass shares the model and rubric but removes the architecture. Detection rises from keyword presence, through single-pass adequacy reasoning, to bounded per-dimension auditing. The final row is the leaked, pre-fix circular setup.

Assessor	detection	false alarm	localization
keyword baseline	0.182	0.000	0.182
single-pass (ablation)	0.455	0.000	0.455
two-stage ARGUS	1.000	0.000	0.909
(<i>circular, pre-fix</i>)	1.000	0.000	1.000

`spillover_contamination`) by the *absence* of their vocabulary, but beyond those detection is only 0.182 (Figure 3B). Two failure modes explain it: *cross-section vocabulary bleed*, where deleting the placebo section does not help because another section also mentions “placebo,” and, more fundamentally, blindness to *every* commission flaw, where flawed-but-plausible evidence is present in the expected topic vocabulary. Keyword presence is not evidence adequacy.

Reasoning over adequacy closes the gap. Swapping *only* the assessment policy (extraction unchanged, so both see the same evidence) to an LLM that judges adequacy raises detection from 0.182 to 1.000 with no false alarms and 0.909 localization. The model reads flawed-but-natural prose the keyword scorer cannot: for the injected anticipation flaw it returns *high/flawed*, reasoning that “outcomes began changing between the policy announcement and rollout,” with no sentinel keyword present. This is the empirical core of the design: placing agency where reasoning is required.

The architecture, not the model, does the work. The single-pass ablation isolates the bounded design. On the flaw-injection benchmark it detects only 0.455 of flaws (false alarm 0.000, localization 0.455), against 1.000/0.000/0.909 for the two-stage assessor: judging eleven dimensions from the whole paper at once misses more than half the planted flaws that per-dimension retrieval isolates. On the 25 papers common to both runs (275 cells) the contrast sharpens. Single-pass almost never abstains: its unknown share is 0.4% versus 38.9% for the two-stage pipeline. The cells the bounded pipeline marks unknown become con-

fidest judgements under single-pass, mostly *low* (1.8% → 14.9%) or *medium* (14.2% → 43.6%), while the high-risk share is comparable (41.1% vs 45.1%). Without a gate the model commits a substantive risk for nearly every dimension instead of reporting that it could not ground the evidence. The bounded design thus recovers flaws a single pass misses and turns retrieval failure into a visible, auditable signal: the distinction the rest of the section relies on.

Scope. These are synthetic fixtures with few dimensions; 1.000 is a proof-of-concept on clearly injected flaws, not a solved task. The sharper questions are whether the behaviour survives subtler, more varied flaws, and then real published papers.

5.2 Robustness: a larger, harder variant benchmark

The eleven flaws above are deliberately clear-cut. To test whether the result survives subtler and more varied flaws, we built a second benchmark of 33 flaw variants: three per dimension (two *commission* rewrites and one *omission*), each verified to contain none of the keyword detector’s sentinel phrases. We audit this set with the two-stage ARGUS assessor (gpt-4o) over three runs at temperature 0; verdicts are near-deterministic (32/33 identical across runs). Detection is 0.73–0.76 and localization 0.64–0.70, below the saturated 1.000 on the eleven clear flaws, which is the point: a harder set separates what the assessor can and cannot do.

The split by flaw type is the informative result (Table 2). The assessor catches 0.86–0.91 of *commission* flaws (flawed-but-present evidence, the case adequacy reasoning targets) but only 0.45 of *omission* flaws. The omission misses are not confident errors: the six missed variants all abstain to *unknown* rather than assert low risk, because deleting the supporting section makes retrieval fail. This reproduces, under controlled injection, the same *evidence-grounding* bottleneck we find on real papers (§5.3): the pipeline reasons well over evidence it can retrieve and abstains when it cannot. The only false-alarm signal (0.09, identical across runs) is concentrated in *inference*, where the clean fixture’s section is itself flagged, the same over-severity we quantify against the human gold (§5.5). The “no false alarms” result in Table 1 is on the original, clearer eleven-flaw set; this harder benchmark surfaces the inference over-flag.

Table 2: Harder 33-variant benchmark (gpt-4o, mean of three runs at temperature 0; near-deterministic, 32/33 verdicts identical). Adequacy reasoning catches *commission* flaws; *omission* misses are abstentions to *unknown*, marking the evidence-grounding bottleneck rather than wrong judgements.

variant type	<i>n</i>	detection	localization
commission	22	0.89	0.82
omission	11	0.45	0.33
all	33	0.75	0.66

5.3 From synthetic fixtures to real papers: a relocated bottleneck

We next audit a corpus of 27 published DID papers. There is no injected ground truth; we report the raw per-dimension risk distribution of each assessor (Figure 5). The keyword baseline is permissive: 91% of its judgements are *low*. It sees topic vocabulary, declares the dimension supported, and flags almost nothing. The two-stage LLM is differentiated, and its most informative output is that ~39% of judgements are *unknown*: the relevance gate could not surface evidence, so the system says so rather than over-flagging. Two regimes appear: dimensions whose discussion is diffuse or lives in figures the text retrieval cannot reach (parallel trends, no anticipation, placebo) are dominated by *unknown*, while dimensions where a section is retrieved (sample period, concurrent policies, SUTVA) are judged inadequate. The finding is methodological: moving to real papers is bottlenecked not by causal reasoning but by *evidence grounding*. The design therefore reports retrieval quality and an explicit *unknown* state distinct from a substantive high-risk judgement.

Judgements are grounded, not pattern-matched. On a trade-and-wages paper ARGUS retrieves the employment-results section and flags *inference* as high because “the paper only mentions heteroskedastic-consistent standard errors and does not provide information on clustering or the number of clusters”: a specific econometric observation, not a keyword hit; elsewhere it distinguishes a *related but different* reported check (an alternative-specification section) from the placebo test an assumption actually requires. Roughly 45% of its judgements on these top-journal papers are *high*; without dimension-level expert labels we cannot yet say whether this is warranted or over-strict where retrieval succeeds. We claim the

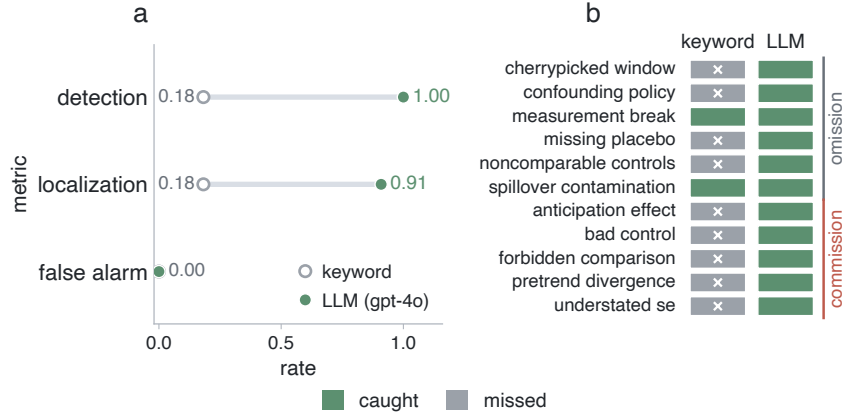


Figure 3: Keyword baseline vs. LLM evidence-adequacy assessor ($n=11$ flaws, realistic injection). **(a)** Swapping *only* the assessment policy (same retrieved evidence) recovers detection $0.182 \rightarrow 1.000$ and localization $0.182 \rightarrow 0.909$; false alarm 0. **(b)** Per-flaw caught/missed: the keyword baseline is *blind to every commission-type flaw*, all of which the LLM catches.

contrast in behaviour and the localization of the grounding bottleneck, not the correctness of any individual flag.

5.4 A calibrated human gold (pilot)

To test whether ARGUS’s flags agree with expert judgement, we built a small expert gold: two annotators independently labelled risk on 5 DID papers $\times$ 11 dimensions (55 cells). Round-1 agreement was low on the raw 3-level scale (Cohen’s $\kappa = 0.17$), but the disagreement is almost entirely ordinal (weighted $\kappa = 0.31$; 96% of disagreements are a single severity step, so annotators almost never reverse direction). A calibration round adjudicated the 27 disagreements: 16 resolved to a single label once the low/medium boundary was anchored, and 11 were flagged intrinsically ambiguous. Those ambiguous cells are not random: they concentrate on *non-canonical DID designs* (continuous-shock, event-study) where a dimension like staggered timing is not applicable as conventionally defined. A real share of “expert disagreement” is thus *design applicability*, not severity; on those cells even experts abstain. The adjudicated gold (55 cells) is low 29, medium 24, high 2. This is a pilot ($n=5$, two annotators, one adjudication round), read as preliminary.

5.5 ARGUS versus the human gold

We use this gold as a *diagnostic pilot*, not a validation of accuracy: with only two high-risk gold cells across five papers it cannot establish agreement with experts, but it is enough to localize how ARGUS errs (Table 3, Figure 4). The er-

rors are systematic, not random. ARGUS abstains on 22/55 cells, and when it answers its errors are almost entirely one-directional: 28/33 answered cells are more severe than the adjudicated gold, and none are less severe. Human gold contains only two high-risk cells, whereas ARGUS assigns high risk to 24. Because the pilot gold has so few high labels, high precision and recall are noisy; the robust conclusion is that the current assessor is over-sensitive to missing or partial evidence and is best read as a *high-sensitivity screening tool* for prioritizing human review.

5.6 Calibration: from diagnosis to improvement

The gold comparison also points to a fix. An error analysis of the over-severe cells is sharp: of ARGUS’s over-severe *high* judgements, 23/24 have `evidence_status=missing` and 20/24 have `retrieval_quality=weak`. ARGUS treats “weak retrieval found nothing” as a substantive *high*, conflating a *retrieval failure* with the paper lacking the evidence, which the human, reading the full PDF, resolves to low/medium. This is a fixable calibration target.

A deterministic calibration layer over fields the LLM already produces cuts over-severity while holding high-risk detection unchanged on the pilot’s two high cells; the dominant rule demotes a weak-retrieval *high* to *medium* (keep coverage) or *unknown* (abstain). Keeping coverage raises exact agreement from 0.15 to 0.45; abstaining reaches weighted $\kappa = 0.25$ but leaves 13 answered cells (Appendix, Table 4, Figure 6). Calibration

corrects severity but cannot recover missing evidence (correct *coverage* needs better grounding), so the human gold does more than evaluate ARGUS; it calibrates it.

6 Discussion

What ARGUS measures. A measurement is only as good as the link between the construct and the task that operationalizes it (Jacobs and Wallach, 2021; Raji et al., 2021). ARGUS audits whether a climate-policy study’s identification assumptions are *adequately supported by its reported evidence*: reported-evidence adequacy is an operational proxy for identification credibility, not causal validity itself. The two can diverge (a study may report every diagnostic yet still be invalid, or omit a check yet be sound), so ARGUS surfaces risk for an expert to weigh and makes no ruling on the effect itself.

Why climate policy. The identification challenges above (staggered rollouts, spillovers, overlapping regulations, anticipation, noisy emissions measurement) stress several assumptions at once, and climate-policy estimates feed directly into governance and resource allocation, so a confidently wrong credibility judgement is costly here; ARGUS therefore abstains rather than guesses when it cannot ground the evidence.

Evidence grounding is the binding constraint. A further gap separates the construct from the task: retrieving the relevant evidence. Our real-paper runs show this, and not the reasoning, is the limiting factor. A reasoning assessor handed off-target evidence over-flags every dimension, so ARGUS reports a *retrieval-quality* status and an explicit *unknown* risk (“the system could not find the evidence”) distinct from a substantive high-risk judgement (“the paper lacks it”). This is why the pipeline separates retrieval, relevance filtering, and adequacy assessment.

Threats to validity. Flaw injection certifies an auditor against the *injected* threat on a single dimension, not against un-injected threats, and it inherits the flaw taxonomy’s blind spots, so we treat detection and localization as necessary but not sufficient, pairing them with the false-alarm rate and the human-gold comparison. Further limitations are stated below.

Future work. Four directions follow from §5: (1) harden the reasoning assessor on subtler flaws, studying calibration, cross-model robustness, and cited-evidence traceability under the same flaw-injection protocol; (2) collect *dimension-level* expert annotation (the real corpus is labelled per method/claim, not per assumption) to test whether ARGUS’s judgements match expert opinion; (3) run human-expert studies of whether ARGUS reports actually help (trust, time, errors avoided), since the system assists rather than replaces expert judgement; and (4) extend beyond the China emissions-trading case to broader environmental and empirical-economics settings, testing how far the rubric-plus-injection methodology generalizes.

7 Conclusion

The right target for AI assistance in empirical climate-policy research is not the unobservable causal effect but the *identification credibility* of the study that estimates it. ARGUS operationalizes this stance: a bounded-agentic pipeline decomposes identification into auditable dimensions, gathers and assesses in-paper evidence, and localizes risk in a transparent, human-reviewable report, while confining model agency to two scoped, traceable sub-tasks. We validate such auditors with synthetic flaw injection, which supplies reproducible local ground truth without access to the true effect. Once the leakage that made this evaluation circular is removed, a keyword baseline is blind to flawed-but-plausible evidence (0.182 detection), while a model that judges evidence adequacy recovers it (1.000, no false alarms). A human-gold diagnostic pilot further shows that expert labels can do more than evaluate the auditor: they expose a systematic over-severity bias and calibrate the mapping from evidence gaps to audit risk. Scaling this from synthetic fixtures to a real, expert-annotated corpus with humans in the loop remains the central open challenge. Auditing identification credibility is, we argue, both a tractable and a responsible role for AI in empirical environmental-policy research.

Limitations

Our human gold is pilot-scale: 5 papers and 55 cells with two annotators and a single adjudication round, so the expert comparison is a preliminary signal rather than a calibrated

benchmark. The current scope is confined to difference-in-differences identification; whether the rubric-plus-injection methodology transfers to other quasi-experimental designs is untested. Evidence grounding, not reasoning, is the binding constraint: when retrieval returns off-target passages the assessor cannot judge adequacy, which is why ARGUS emits an explicit unknown state rather than a substantive verdict. Finally, ARGUS is an auditor, not an oracle: it screens whether reported evidence supports a study’s identification assumptions and does not adjudicate the true causal effect, which is never observed. Its outputs are meant to surface risk for expert review, not to replace it.

References

- Joshua D. Angrist and Jörn-Steffen Pischke. 2010. The credibility revolution in empirical economics: How better research design is taking the con out of econometrics. *Journal of Economic Perspectives*, 24(2):3–30.
- Susan Athey and Guido W. Imbens. 2017. The state of applied econometrics: Causality and policy evaluation. *Journal of Economic Perspectives*, 31(2):3–32.
- Marianne Bertrand, Esther Duflo, and Sendhil Mullainathan. 2004. How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics*, 119(1):249–275.
- Rotem Botvinik-Nezer, Felix Holzmeister, Colin F. Camerer, et al. 2020. Variability in the analysis of a single neuroimaging dataset by many teams. *Nature*, 582(7810):84–88.
- Abel Brodeur, Nikolai Cook, and Anthony Heyes. 2020. Methods matter: P-hacking and publication bias in causal analysis in economics. *American Economic Review*, 110(11):3634–3660.
- Brantly Callaway and Pedro H. C. Sant’Anna. 2021. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.
- Clément de Chaisemartin and Xavier D’Haultfœuille. 2020. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996.
- Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2021. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92.
- Andrew Gelman and Eric Loken. 2014. The statistical crisis in science. *American Scientist*, 102(6):460–465.
- Andrew Goodman-Bacon. 2021. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.
- John P. A. Ioannidis. 2005. Why most published research findings are false. *PLoS Medicine*, 2(8):e124.
- Abigail Z. Jacobs and Hanna Wallach. 2021. Measurement and fairness. In *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*.
- Zhijing Jin, Yuen Chen, Felix Leeb, Luigi Gresele, Ojasv Kamal, Zhiheng Lyu, Kevin Blin, Fernando Gonzalez, Max Kleiman-Weiner, Mrinmaya Sachan, and Bernhard Schölkopf. 2023. CLadder: Assessing causal reasoning in language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Zhijing Jin, Jiarui Liu, Zhiheng Lyu, Spencer Poff, Mrinmaya Sachan, Rada Mihalcea, Mona Diab, and Bernhard Schölkopf. 2024. Can large language models infer causation from correlation? In *International Conference on Learning Representations (ICLR)*.
- Emre Kiciman, Robert Ness, Amit Sharma, and Chenhao Tan. 2024. Causal reasoning and large language models: Opening a new frontier for causality. *Transactions on Machine Learning Research (TMLR)*.
- Douwe Kiela, Max Bartolo, Yixin Nie, et al. 2021. Dynabench: Rethinking benchmarking in NLP. In *NAACL-HLT*.
- Edward E. Leamer. 1983. Let’s take the con out of econometrics. *American Economic Review*, 73(1):31–43.
- Donggyu Lee, Hyeok Yun, Meeyoung Cha, Sungwon Park, Sangyoon Park, and Jihee Kim. 2025. EconCausal: A context-aware economic reasoning benchmark for large language models. Preprint, arXiv:2510.07231. Preprint, arXiv:2510.07231.
- Percy Liang, Rishi Bommasani, Tony Lee, et al. 2023. Holistic evaluation of language models. *Transactions on Machine Learning Research (TMLR)*.
- Weixin Liang, Yuhui Zhang, Hancheng Cao, Binglu Wang, Daisy Yi Ding, Xinyu Yang, Kailas Vodrahalli, Siyu He, Daniel Scott Smith, Yian Yin, Daniel A. McFarland, and James Zou. 2024. Can large language models provide useful feedback on research papers? a large-scale empirical analysis. *NEJM AI*, 1(8).
- Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. 2024. The AI scientist: Towards fully automated open-ended scientific discovery. Preprint, arXiv:2408.06292. Preprint, arXiv:2408.06292.
- Inioluwa Deborah Raji, Emily M. Bender, Amanda-lynn Paullada, Emily Denton, and Alex Hanna. 2021. AI and the everything in the whole wide

world benchmark. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

Ashesh Rambachan and Jonathan Roth. 2023. A more credible approach to parallel trends. *Review of Economic Studies*, 90(5):2555–2591.

Jonathan Roth. 2022. Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3):305–322.

Jonathan Roth, Pedro H. C. Sant’Anna, Alyssa Bilinski, and John Poe. 2023. What’s trending in difference-in-differences? a synthesis of the recent econometrics literature. *Journal of Econometrics*, 235(2):2218–2244.

Ayush Sawarni, Jiyuan Tan, and Vasilis Syrgkanis. 2026. [CausalReasoningBenchmark: A real-world benchmark for disentangled evaluation of causal identification and estimation](#). Preprint, arXiv:2602.20571. *Preprint*, arXiv:2602.20571.

Raphael Silberzahn, Eric L. Uhlmann, Daniel P. Martin, et al. 2018. Many analysts, one data set: Making transparent how variations in analytic choices affect results. *Advances in Methods and Practices in Psychological Science*, 1(3):337–356.

Joseph P. Simmons, Leif D. Nelson, and Uri Simonsohn. 2011. False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11):1359–1366.

Liyang Sun and Sarah Abraham. 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.

Linying Yang, Vik Shirvaikar, Oscar Clivio, and Fabian Falck. 2024. [A critical review of causal reasoning benchmarks for large language models](#). Preprint, arXiv:2407.08029. *Preprint*, arXiv:2407.08029.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.

Matej Zečević, Moritz Willig, Devendra Singh Dhami, and Kristian Kersting. 2023. Causal parrots: Large language models may talk causality but are not causal. *Transactions on Machine Learning Research (TMLR)*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhonghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

Yu Zhou, Xingyu Wu, Beicheng Huang, Jibin Wu, Liang Feng, and Kay Chen Tan. 2024. [CausalBench: A comprehensive benchmark for causal learning capability of large language models](#). Preprint, arXiv:2404.06349. *Preprint*, arXiv:2404.06349.

A Domain Motivation and Design Rationale

A.1 Why climate-policy DID is a demanding audit setting

Climate-policy evaluations present several recurring identification challenges. First, policy announcement, legal adoption, and actual implementation may occur at different dates, creating scope for anticipation. Second, carbon markets and environmental regulations are commonly introduced at different times across jurisdictions, making heterogeneous and staggered treatment effects central rather than exceptional. Third, emissions, investment, and production responses can spill across regional borders, challenging no-interference assumptions. Fourth, climate policies frequently overlap with industrial, energy, and air-pollution regulations, complicating the exclusion of concurrent interventions. Fifth, policy effects may emerge with long and heterogeneous lags. Finally, emissions and environmental outcomes are assembled from administrative, modeled, or remotely sensed sources whose measurement properties vary across places and time. These features do not make DID unique to climate policy, but they make the quality of identification evidence especially consequential for policy interpretation and comparison.

A.2 Identification auditing and research reliability

ARGUS addresses one component of a broader research-reliability problem: published conclusions can vary with specification choices, undisclosed analytic flexibility, and researcher degrees of freedom (Leamer, 1983; Ioannidis, 2005; Simmons et al., 2011; Gelman and Loken, 2014; Silberzahn et al., 2018; Brodeur et al., 2020; Botvinik-Nezer et al., 2020). This is the identification-credibility component of the broader research-design or “credibility revolution” program in applied econometrics (Angrist and Pischke, 2010; Athey and Imbens, 2017), which increasingly treats an assumption like parallel trends as a matter of degree rather than a binary (Ram-

Table 3: ARGUS-LLM vs adjudicated human gold (55 cells, 5 papers) under two unknown policies: coverage (drop abstentions) and strict (unknown counts as mismatch). Pilot-scale: only 2 gold cells are *high*; the robust signal is the one-directional over-severity and the abstention rate.

unknown policy	n	exact	κ	wt. κ
exclude (coverage)	33	0.152	0.019	0.044
mismatch (strict)	55	0.091	0.002	–

Table 4: Deterministic calibration vs the human gold (55 cells), two policies for weak-retrieval *high*: demote to *medium* (keep coverage) or abstain to unknown. Both cut over-severity and preserve high recall. Pilot scale: only 2 gold cells are *high*.

metric	before	demote→med	abstain→unk
over-severe (answ.)	28	18	6
high prec. / rec.	0.04 / 0.50	0.50 / 0.50	0.50 / 0.50
exact agr. (answ.)	0.15	0.45	0.54
wt. κ (answ.)	0.06	0.13	0.25
answered / unk	33 / 22	33 / 22	13 / 42

bachan and Roth, 2023; Roth, 2022). ARGUS does not attempt to audit every component of an empirical paper; it focuses on the identification argument that licenses a causal interpretation.

A.3 Why ARGUS uses bounded agency

Open-ended agents may choose different tools and reasoning paths across runs, making their outputs difficult to reproduce and compare. Large language models are increasingly used for evaluative and reviewing tasks (Zheng et al., 2023; Liang et al., 2024; Lu et al., 2024), but as open-ended agents their reasoning traces are hard to audit. ARGUS instead fixes the pipeline order in code and confines model-driven decisions to evidence retrieval and adequacy assessment. Each model call operates under a fixed tool set, a hard step budget, and a logged trace. This design trades autonomy for reproducibility, inspectability, and controlled evaluation under flaw injection.

B Supplementary Figures and Tables

This appendix collects supporting material referenced from the main text: the per-dimension risk distribution on the 27-paper real corpus (Figure 5), and the deterministic calibration results against the adjudicated human gold (Table 4, Figure 6), and the agreement metrics for ARGUS versus the human gold (Table 3).

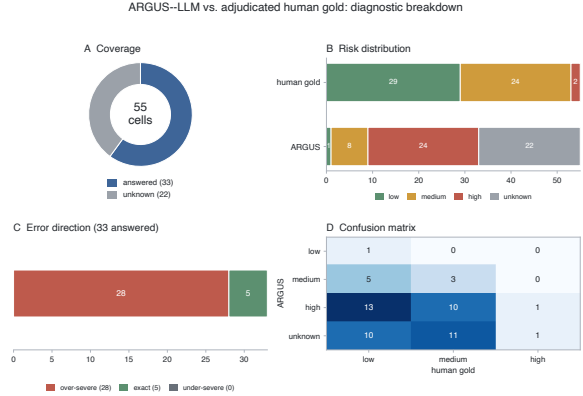


Figure 4: ARGUS-LLM versus adjudicated human gold on 55 dimension-level cells (5 papers). (a) coverage, separating answered judgments from abstentions; (b) severity distributions of human gold and ARGUS; (c) error direction among the 33 answered cells; (d) the ARGUS-by-gold confusion matrix. The dominant signal is one-directional over-severity: ARGUS assigns higher risk than gold in 28/33 answered cases and lower risk in none.

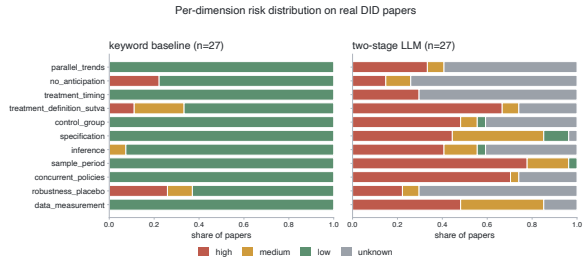


Figure 5: Per-dimension risk distribution on 27 real DID papers. The keyword baseline (left) is almost entirely *low*; the two-stage LLM (right) is differentiated, with a large unknown share that marks retrieval failure rather than over-flagging.

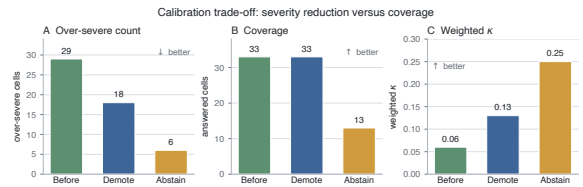


Figure 6: **Calibration trade-off: severity reduction versus coverage** against the adjudicated human gold. Demoting weak-retrieval *high*s to *medium* preserves coverage (33 answered) while cutting over-severe judgments from 29 to 18; abstaining gives the strongest correction (over-severe 6, weighted κ 0.25) but leaves only 13 answered cells. Calibration corrects severity bias; the remaining bottleneck is evidence grounding, not severity thresholding alone.